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Manufacturing cost estimation based on the machining process and deep-learning method

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ABSTRACT

With the extensive application of mass customization, fast and accurate responses to customer inquiries can not only improve the competitive advantage of an enterprise but also reduce the cost of parts at the design stage. Most cost estimation methods establish a regression relationship between features and cost based on the processing features of parts. Traditional methods, however, encounter certain problems in feature recognition, such as the inaccurate recognition of processing features and low efficiency. Deep-learning methods have the ability to automatically learn complex high-level data features from a large amount of data, which are studied to recognize processing features and estimate the cost of parts. First, this study proposes a novel three-dimensional (3D) convolutional neural network (CNN) part-feature recognition method to achieve highly accurate feature recognition. Furthermore, an innovative method of using the quantity to express the identified features and establishing the relationship between them and cost is proposed. Then, support vector machine and back propagation (BP) neural network methods are employed to establish a regression relationship between the quantity and cost. Finally, in comparison with the mean absolute percentage error values, the BP neural network yields a more accurate estimation, which has considerable application potential.

1. Introduction

In recent years, the competition among manufacturers of parts is not only in terms of quality and technology, but also with regard to the goal of delivering the highest quality products within the shortest delivery time. With the wide application of customization, it is important for enterprises to quickly and accurately respond to customer inquiry. The accurate cost estimation of parts can aid designers to easily understand the manufacturing cost of products in the product design stage and timely improve the design to reduce production cost. Although the estimated cost of parts only accounts for approximately 5 % of the total cost of product development, it represents 70–80 % of the total cost of parts [1].

Cost estimate is a quantitative approximation of the resource cost required for part processing. It is based on the premise that the company does not extract the production schedule and data from the manufacturing process to estimate product cost, thus indicating a typical regression problem. The typical methods that are used for estimating the manufacturing cost of parts are the multi-attribute fuzzy method [2,3], similarity estimation method [4,5], cost estimation method based on feature recognition techniques [6], and neural network estimation method [7].

The multi-attribute fuzzy method [2] Ref. to the technique of fuzzy multi-attribute effect theory based on product cost information. This method is employed to estimate product cost by determining the upper and lower limits as well as the weight coefficients of cost drivers to analyze the starting point of product cost drivers in the design stage. This method can also reduce the error in product cost estimation. Baykasoğlu et al. [3] provided a new fuzzy multiple-attribute decision-making model for evaluating product pricing strategies. Possible interactions and interdependencies among hierarchically structured criteria are taken into consideration. The model is implemented in a Turkish software company.

The similarity estimation method [4] is employed to approximate product cost based on the similarity relationship among products. Through similarity calculations, existing samples similar to the target samples are retrieved in the sample bank, and the cost of target samples is determined by cost calculation. Relich and Pawlewski, [5] presented the use of artificial neural network to calculate attribute weights in the case-based reasoning (CBR) approach in the context of case retrieval

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and reuse for the cost estimation of new product development (NPD) projects. Cost estimation is employed to select the most promising portfolio of NPD projects. Thereafter, it is also utilized in the revision phase of the CBR to monitor the NPD process.

The cost estimation method based on feature recognition techniques is employed to estimate the manufacturing cost through various manufacturing steps of a series of parts, such as tooling, labor cost, transportation cost, packaging cost, manufacturing parameters, processing equipment, and loss. This method can accurately estimate the manufacturing cost of parts, but it requires a large amount of information, including part design and manufacturing information. Grund and Schmidt [6] presented analytical cost models for spatial light modulators that adapt and expanded some approaches to a generic model by automatically calculating the unit costs of parts. These costs, which are determined by statistical methods, include the build height of a part or the capacity utilization of a build job.

The neural network estimation method is based on a large amount of cost-related data as the neural network input and the training data cost as the output. Through the neural network regression operation, the relationship between input and output is obtained. Machine learning techniques were used by Loyer et al. [7] to rapidly estimate the cost of jet engine components. In their research, they found that learning appears to be an effective, affordable, accurate, and scalable technique to determine the cost of mechanical parts.

In addition, in many parts manufacturers, machining time is used to estimate part cost [8]. Cost is proportional to machining time, which includes operational time and non-operational time. Operational time includes rough cutting time and finish cutting time. This method can accurately estimate the cost of certain parts. However, for some rare or combined features, as well as some special features, owing to the lack of feature data, the estimation error is large. Deng and Yeh, [9] applied a least-squares support vector machine method to solve the problem of estimating the manufacturing cost for airframe structural projects. This research considered structural weight and manufacturing complexity as the main factors in determining the manufacturing labor time. Luiza et al. [10] proposed a product time and cost estimation method taking account market dynamics and manufacturing system competitiveness, which is based on behavioral modeling using online unsupervised learning.

Moreover, there are some quantitative methods that include operation-based [11] and breakdown approaches [12]. The operation-based approach estimates the cost mainly according to the type of operation, where the cost includes factory cost, material cost, operation time, and other factors. The breakdown approach partitions manufacturing costs into cost elements. An estimation is applied to each cost element. The estimated manufacturing cost is a sum of all cost elements incurred during the production cycle. Lee and Kulvatunyou, [13] described a distributed manufacturing situation and a cost breakdown framework to estimate cost in the early design stage, which is difficult owing to insufficient information.

Other regression techniques, such as analysis estimation method [14], functional cost estimation method [15], learning curvature estimation method [16], nonparametric cost estimation method [17], and activity-based cost method [18] are also frequently used. Table 1 lists the current commonly used methods for part cost estimation. These methods are based on the regression relationship between input quantity and cost, and most are based on the processing features of parts. It can be concluded that the processing feature recognition performs a crucial function in the cost estimation of parts.

The feature recognition of parts has been developed for half a century. To date, the typically employed methods of manufacturing feature recognition are based on graph, trace, convex, and cell.

The method of feature recognition based on graphs represents the geometric topological structure of part manufacturing features through the graph of faces and edges and to form attributed adjacency graphs (AAGs) by analyzing the convexity of faces or edges on all the nodes of faces and edges of parts. The advantage of this method is that researchers can incorporate different face node attributes according to different research directions. Moreover, this method is highly accurate in obtaining independent part features. However, it cannot accurately identify the intersection feature because of the loss or imperfection of the feature surface or edge caused by the intersection feature. Wang et al. [19] proposed an oriented feature extraction and recognition approach for concave–convex-mixed interacting features. The method first directionally extracts predefined features according to the rules generated from the attributed adjacency graph (AAG), and the subfeatures in an interacting feature are associated by means of hints and organized as a feature tree. This method can effectively recognize some intersecting features and improve the recognition speed. The method of constructing the attribute adjacency matrix by the attribute adjacency graph to complete the matching of feature subgraphs is confronted with the nondeterministic polynomial time problem of computers, making this method time-consuming and inefficient. In order to solve this problem, some scholars have employed the graph trace method and neural networks to identify manufacturing features and have achieved good results [20].

The trace-based feature recognition method was proposed by Vandenbrande and Requicha, [21]. This method is based on the analysis of the information of manufacturing features in the part model. When the manufacturing features intersect, the complete boundary pattern of features is broken, but the traces of the manufacturing features of parts remain in the part model. The advantage of this method is that it can accurately identify the intersection features of parts. Its disadvantage, however, is that it has difficulty in identifying the traces that correspond to the manufacturing features of parts and in finding the appropriate traces for complex features. At the same time, the feature traces depend on the specific feature types. Li et al. [22] proposed a symmetry property definition relative to the edges of solid models to support hint-based reasoning and processing. This symmetry property is based on conventional edge convexity, which is an extension of the feature hint theory.

Convex hull and cell-based decompositions are two approaches based on decomposing the input model into a set of intermediate volumes and manipulating the volumes in order to produce features. The method of cell-based decomposition enumerates parts and blanks in space so that parts can be divided into blanks, parts, and half parts. According to the types and rules of cells, the cutting bodies of parts can be searched out, and manufacturing features can be obtained [23,24]. In the decomposition process, the global effect of local geometry is encountered. This makes the method less efficient and requires more

Table 1
Comparison of various estimation methods.

Models	Method name	Base Method	Applicable stage	Method difficulty
Regression model	Similarity estimation method	Similarity calculation	Completion phase	Difficult
	Multi-attribute fuzzy method	Cost drivers		
	Feature-based and process-based estimation method	Processing feature recognition		
	Regression analysis estimation		Early stage of design	General difficulties
	Function cost estimation			
	Neural network estimation method			

computations.

The feature recognition process based on convex decomposition involves the following: first, the solid model of the part is decomposed into a geometry composed of convex bodies; thereafter, the convex body geometry is recombined according to certain rules for forming the feature set. The techniques employed in the process are alternating sum of volumes and alternating sum of volumes with partition methods [25,26]. The convex body is decomposed into a tree structure by Boolean operation. The subsequent node represents the part itself, and the leaf node represents the convex body on the root node. It is necessary to ascertain whether or not the leaf node corresponds to a feature. If it does not, this node is combined with other leaf nodes until a feature is formed. In theory, this method can identify the intersection features; however, some of the identified features may be incorrect, and a considerable amount of calculation is involved in the combination.

There are also other feature recognition methods: those based on logic rules and expert system [27], ray distance method [28], and convolution neural network method [29].

The key to the estimation of part manufacturing cost is recognizing the part-processing features and the regression relationship among the costs established from these features. In our research, we focus on how to identify the processing features of parts accurately and quickly (in Sections 2 and 3), and then establish an accurate regression relationship between features and cost (Sections 4 and 5). The contribution of this work is to make full use of deep-learning technology to realize the recognition of processing features of parts, and to establish a regression relationship between features and cost. Specifically, Section 2 presents the separation of part features by the minimal subgraph method. In Section 3, the 3D CNN method is proposed to achieve feature recognition. In order to describe the processing features of parts in terms of quantity and establish the relationship between features and cost, this study calculates the feature quantity corresponding to each feature, achieves data dimensionality reduction with the principal component analysis (PCA) method, and obtains the feature data as explained in Section 4. Sections 5 and 6 introduce the support vector machine (SVM) and BP neural network methods. The use of LibLinear, LibSVM, and BP network to establish the linear and nonlinear regression relationships between these characteristic data and cost. Finally, this paper compares these three methods, according to the mean absolute percentage error (*MAPE*) value to evaluate the estimated effect.

2. Part-processing feature splitting

2.1. Part-processing feature judgment

The machining of parts is the creation of a model by cutting, milling, boring, and other machining methods to remove the designated redundant materials according to the drawing of the part. In removing the redundant material, concave and convex shapes of the part blank model surface and edge are changed, making the part-processing features to have concave and convex edges. According to the geometric topological information of the part model, this study employs the method of judging the concave–convex connection between adjacent nodes to derive the AAG of the part and thereafter obtain the minimum subgraph of the concave edge connection of the part to judge the processing features.

The method of judging the concavity and convexity of the straight line edge involves the following. As shown in Fig. 1, the external normal vectors of adjacent surfaces, F_1 and F_2, are N_1 and N_2, respectively; F_1 is taken as the base surface, and each edge of F_1 constitutes a closed ring whose edge is e. The direction of the ring is determined by the outer normal vector, N_1, of the plane according to the right-hand spiral rule. The direction vector N_e of edge e is consistent with the direction of the ring. The direction of vector N is determined according to $N = N_e \times N_2$. If the angle between N and N_1 satisfies the condition $0 \leq \theta \leq 90°$, then this is a concave edge; if $90° \leq \theta \leq 180°$, then the edge is convex.

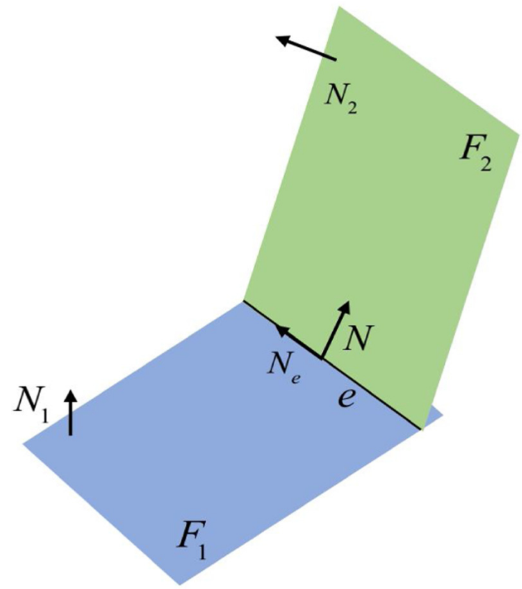


Fig. 1. Concave judgment of line edge.

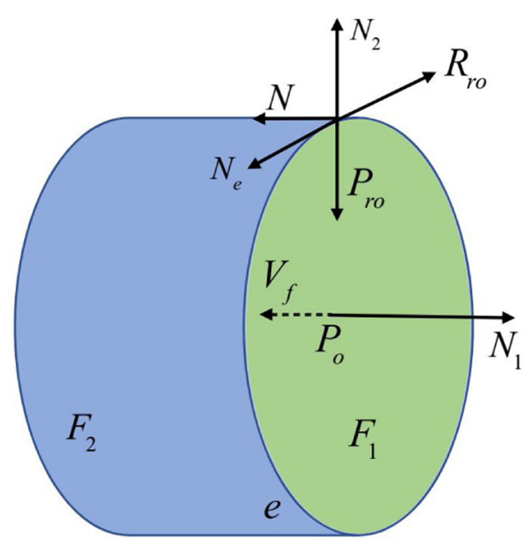


Fig. 2. Concave judgment of arc edge.

The method of judging the concave–convex edge of the arc shown in Fig. 2, the direction of vector N_e is determined according to the normal vector, N_1, outside the adjacent surface, F_1, and the right-hand helix rule. By considering any point, P_r, in the arc, the direction vector, P_{ro}, from point P_r to the center (P_o) is obtained. The direction vector of the first direction entity of the circular curve defined by the geometry corresponding to the cylinder face of edge e is taken as V_f. The tangent vector direction at point P is determined according to P_r by $R_{ro} = P_{ro} \times V_f$. If the cylindrical surface is the outer surface, then the direction of N_2 is determined according to $N_2 = R_{ro} \times V_f$; if the cylindrical surface is an internal surface, then the direction of N_2 is determined according to $N_2 = V_f \times R_{ro}$. Finally, the direction vector, N, is determined according to. If the angle between N and N_1 satisfies the condition $0 \leq \theta \leq 90°$, then this is a concave edge; if $90° \leq \theta \leq 180°$, then the edge is convex.

In this study, a step file is read to obtain all geometric models of the part, and all face and edge information in the model is stored in the form of geometry. The faces are thereafter numbered and expressed in the form of nodes. To obtain the AAG, the faces are connected to the intersecting edges with lines, and the concavity of the connection

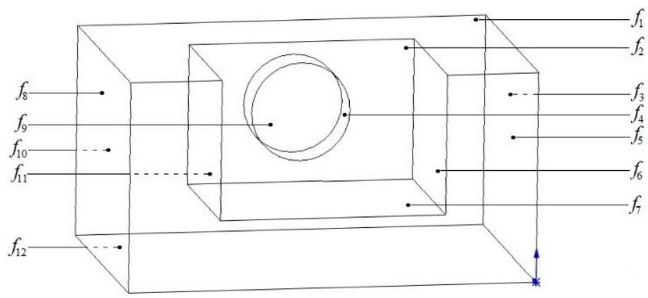


Fig. 3. 3D part with hole and groove.

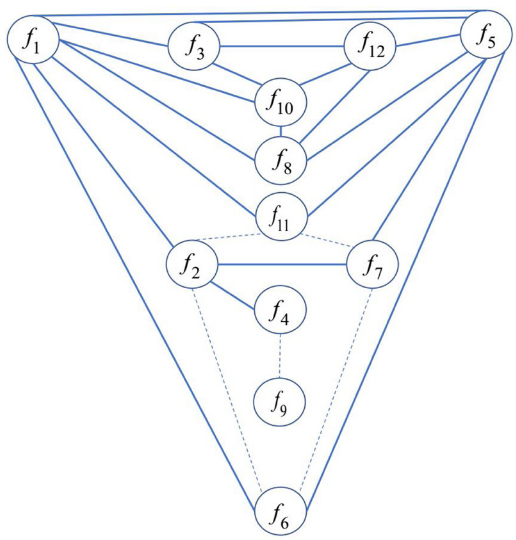
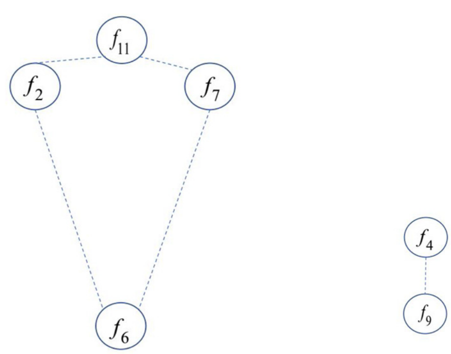


Fig. 4. AAG diagram of part.



(a) Minimum subgraph of groove (b) Minimum subgraph of hole

Fig. 5. A part with processing features.
(a) Minimum subgraph of groove. (b) Minimum subgraph of hole.

between two nodes is judged according to the concavity algorithm of the edges. Finally, the convex edge connection is removed, and the concave edge connection is retained to generate the minimum subgraph according to the concavity attribute of the edges. These minimal graphs are the processing feature surfaces of parts. In the 3D part model shown in Fig. 3, the part model contains the processing features of holes and grooves. Fig. 4 shows the AAG diagram of the part, where the solid and dotted lines represent the convex and concave edges of the part, respectively. The minimum subgraph can be generated from the concave convex of the edge. This achieves the separation of the connection of the hole and groove, which are the processing features of the part, as shown in Fig. 5.

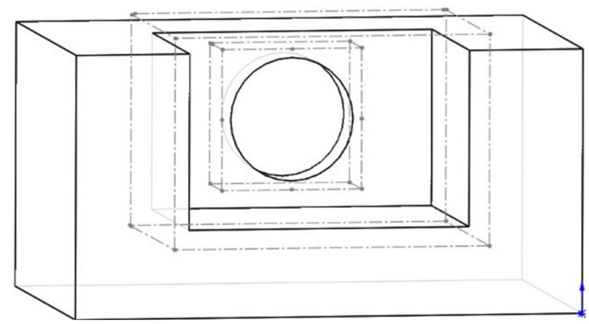


Fig. 6. Binding box of feature surface of a part.

2.2. Bounding box of part-processing features

The minimum subgraph generated from the AAG is only the part's processing feature surface, in order to obtain the entity of the minimum subgraph. In this study, the bounding box method is employed to materialize the processing features of a part without generating new processing features.

The basic idea of the bounding box is the approximate replacement of complex geometric objects with the geometry of slightly larger volumes and simple characteristics (called bounding box). In this study, the minimum graph of parts is obtained by the AAG of parts, and the separation of processing and non-processing features is completed. The processing features of parts are thereafter bound into boxes, as shown in Fig. 6. The bounding-boxization refers to the intersection of the six bounding box planes of the minimal subgraph and the processing features of the part. If the bounding box of the intersection coincides with the outermost part of the intersection, then the part is a concave body. The outermost of the intersection is translated to a positive direction to obtain a larger bounding box. If the outermost part of the intersection is the feature surface of the part, then the part feature is convex. By this means, the part-processing features are separated and are on a hexahedron. Fig. 7 shows the 3D model of separated processing features.

3. 3D CNN feature recognition

The application range of the neural network depth is extended from the two-dimensional model to the three-dimensional model, from the initial two-dimensional image classification to object detection, and the visual research of three-dimensional object model, thus achieving a wider range of applications [30,31]. Voxels can completely describe the space occupation of objects in the form of extremely simple data, which are considerably suitable for existing learning methods. In this study, a method for training the 3D CNN is proposed to voxelate the 3D model of the part and generate the training data as the input data of 3D CNN. After training, the 3D CNN achieves part-processing feature recognition.

3.1. Voxelization

Voxelization transforms the geometric representation of the model into a voxel representation closest to the model. It also generates voxel datasets, which not only contain the surface information of the model, but also describe the internal attributes of the model. The spatial voxels of the model are similar to the two-dimensional pixels of the image, only extending from two-dimensional points to three-dimensional cube elements.

First, the initial 3D model is subdivided by the octree structure until the specified resolution is reached. The voxel data that correspond to the surface elements of the model are obtained with this precision, which is marked as 1, and those of the other non-surface elements are marked as 0. Each row of model elements is thereafter scanned along

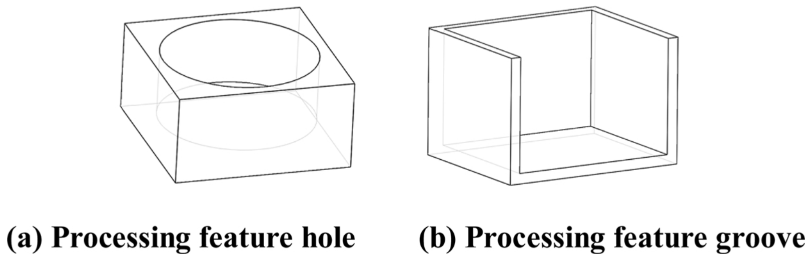


Fig. 7. Processing features of separated part.
(a) Processing feature hole. (b) Processing feature groove.

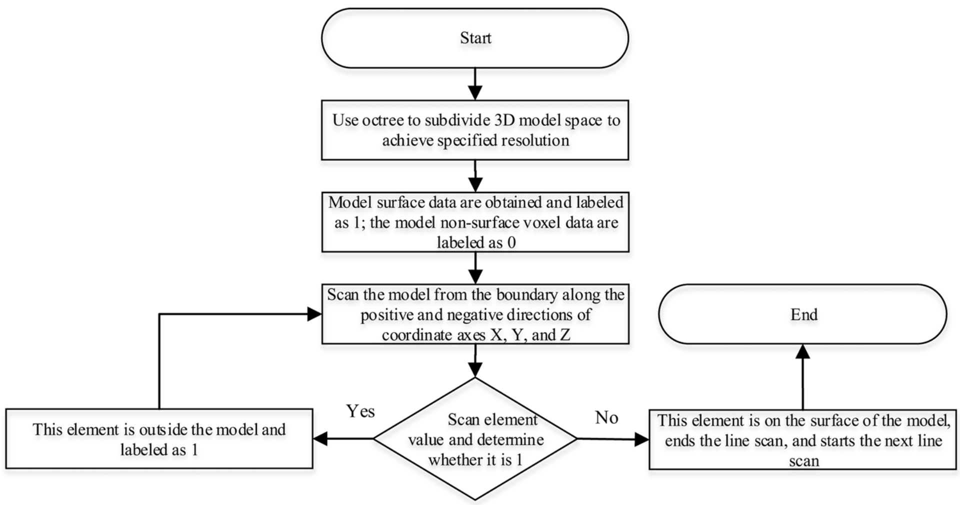


Fig. 8. Voxelization flowchart.

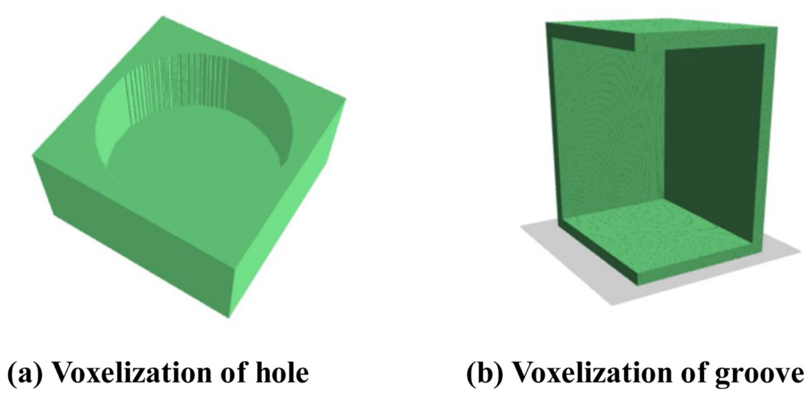


Fig. 9. Processing feature voxelization of separated parts.
(a) Voxelization of hole. (b) Voxelization of groove.

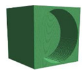
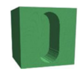
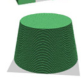
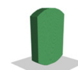
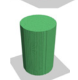
the positive and negative axes of models X, Y, and Z. If the value of the element is 0, then the element is outside the model, and the voxel value is modified to 1; if the value of the element is 1, then the scanning of this row is stopped and thereafter started in the next row. After the positive and negative X, Y, and Z axes are scanned, the surface and external element values of the model are marked as 1, the internal element is marked as 0, and the voxelization of the model is completed. The voxel flow is shown in Fig. 8, and the separated part-processing features are voxelated as shown in Fig. 9.

3.2. Classification of processing features of parts

There are many types of part-processing features because of the various functions and complex structure of parts. Accordingly, in order to facilitate the study of part cost estimation, 14 types of common processing features are selected for this study, as listed in Table 2. These features include all part-processing features in this study, and the method can easily add any processing features.

The study in Ref. [32] shows that the larger the voxel resolution, the more details the voxel data in the model contain, and the more conducive it is for the 3D CNN to learn the features contained in the voxel data. At the same time, however, training will be more difficult. The voxel resolution selected for this study is therefore 128^3. In addition, this study completes the data enhancement in the voxelization process. Through rotation transformation, each model rotates around the *X*, *Y*, and *Z* axes to generate a new model copy. Some voxel data in the part-processing features are listed in Table 3.

Table 2
Processing features for training 3D CNN.

Feature name	Voxel data	Feature name	Voxel data
Non-through hole		Ring	
Bottom groove		Keyway	
Four-sided non-through hole		Three-sided notch	
Five-sided non-through hole		Through hole	
Straight groove		Four-sided through hole	
Platform		Hexahedron	
Decahedron		Cylinder	

4. Characteristic quantity calculation

The processing features and geometric features of parts have an impact on the machining of parts. After identifying the part-processing features, in order to describe the part model in the form of quantity and establish the relationship between quantity and part cost, this study adopts the feature quantity method included in the calculation features, which can describe the part-processing features, such as through hole, diameter, depth, and cylinder. The surface area and quantity are key dimensions that can express the through-hole features. Table 4 lists the feature quantity calculated in the part-processing feature.

4.1. Data dimensionality reduction

The number of features in a part is not constant, and the dimensions of features are different under the same feature conditions. For example, there may be multiple through holes in a part, and the diameter and depth of the through holes may be different, resulting in two-dimensional data with different data lengths. In order to establish the regression relationship between part cost and feature quantity, this study compresses and reduces the dimension of feature quantity data. The PCA (principal component analysis) is a common data analysis method [33,34] that transforms the original data into a set of linear independent representations through linear transformation, which can be employed to extract the main feature components of data.

If it is assumed that dataset X contains m amount of data with p feature quantities, then T is used to represent the vector transposition, and the dataset can be expressed by

standardizing the original matrix, X, and solving the equation.

$$X = (X_1, X_2, \ldots X_M)^T,\ X_i = (x_{i1}, x_{i2}, \ldots x_{ip})^T,\ (i = 1, 2, \ldots, m) \tag{1}$$

Standardizing the original matrix X :

$$X_m \times p = \frac{X - mean(X)}{\sqrt{(std(X))}} \tag{2}$$

Solving the equation:

$$(\lambda I - Y)\alpha = 0 \tag{3}$$

The eigenvalue $Y_{p\times p}$ and the eigenvalue, λ_i, of the covariance matrix are obtained $\alpha_i (i = 1, 2, \ldots, p)$.

The principal component contribution rate and cumulative contribution rate are calculated, parameter k is selected, and the principal component $(Z_1, Z_2, \ldots, Z_k (k \le p))$ is obtained. The analysis object is reduced from P to k dimensions. In this study, $k = 1$ is selected. The contribution rate of the principal component, $Z_i = \alpha_i \times X_{m\times p} (i = 1, 2, \ldots, p)$, is $\lambda_i / \sum_{i=1}^{p} \lambda_i$, $(i = 1, 2, \ldots, p)$. The cumulative contribution rate of principal components is $\sum_{i=1}^{k} \lambda_i / \sum_{i=1}^{p} \lambda_i$, $(i = 1, 2, \ldots, p)$.

4.2. Feature data generation

The generation process of feature data shown in Fig. 7 includes the following. As shown in Fig.10, to obtain the attributes that are used to represent all faces and lines in the 3D model of the part, a complex B-rep data structure is employed to read step AP203 file [35]. The AAG of the part is calculated, and the minimum subgraph of the part in the graph is identified. The intersection of the bounding box and the processing features of the part are also obtained. If the bounding box of the intersection and the outermost part of the intersection coincide, then the part is a concave body. To obtain a larger bounding box, the outermost plane of the intersection is translated to the positive direction. The intersection is thereafter voxelated as the input of 3D CNN. If the outermost part of the intersection is the feature surface of the part, then the intersection is voxelated. The processing features of parts through the trained 3D CNN are recognized, and the feature quantity in the minimum subgraph is calculated according to the recognized part-processing features, as summarized in Table 4. The feature quantity is reduced to 52 unidimensional data by the PCA method and recorded in the SCV file format.

Table 3
Implementing results of some data enhancement with voxel resolution of 128^3.

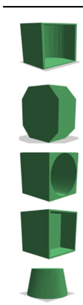
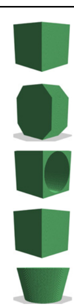
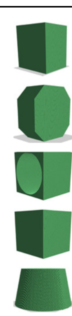
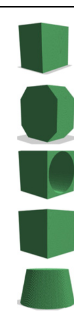
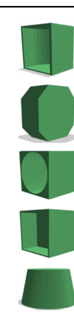
Source model	Rotate 90° around the X-axis	Rotate 180° around the X-axis	Rotate 270° around the X-axis	Rotate 90° around the Y-axis	Rotate 180° around the Y-axis	Rotate 270° around the Y-axis	Rotate 90° around the Z-axis	Rotate 180° around the Z-axis	Rotate 270° around the Z-axis
									

Table 4
Characteristic quantity in part-processing features.

No.	1	2	3	4
Feature quantity	Total feature quantity	Blank volume	Part volume	Total surface area of parts
No.	5	6	7	8
Feature quantity	Number of cylindrical surfaces	Area of cylindrical surfaces	Number of conical faces	Area of cone
No.	9	10	11	12
Feature quantity	Number of spheres	Area of spheres	Diameter of non-through hole	Volume of non-through hole
No.	13	14	15	16
Feature quantity	Number of non-through holes	Diameter of ring	Volume of Ring	Number of rings
No.	17	18	19	20
Feature quantity	Volume of cylinder	Diameter of cylinder	Number of cylinders	Diameter of keyway
No.	21	22	23	24
Feature quantity	Volume of keyway	Number of keyways	Diameter of four-sided non-through hole	Volume of four-sided non-through hole
No.	25	26	27	28
Feature quantity	Number of non-through holes on four sides	Diameter of four-sided through hole	Volume of four-sided through hole	Number of four-sided through holes
No.	29	30	31	32
Feature quantity	Width of three-sided notch	Volume of three-sided notch	Number of three-sided notch	Number of through holes
No.	33	34	35	36
Feature quantity	Diameter of through hole	Volume of through hole	Number of five-sided non-through hole	Volume of five-sided non-through hole
No.	37	38	39	40
Feature quantity	Volume of Five-sided non-through hole	Average diameter of platform	Surface area of platform	Volume of platform
No.	41	42	43	44
Feature quantity	Width of Straight groove	Volume of Straight groove	Number of Straight grooves	Area of Hexahedral surface
No.	45	46	47	48
Feature quantity	Area of Hexahedral surface	Number of Hexahedrons	Depth of Round bottom groove	Volume of Round bottom groove
No.	49	50	51	52
Feature quantity	Number of Round bottom groove	Area of Decahedron surface	Volume of Decahedron	Number of Decahedrons

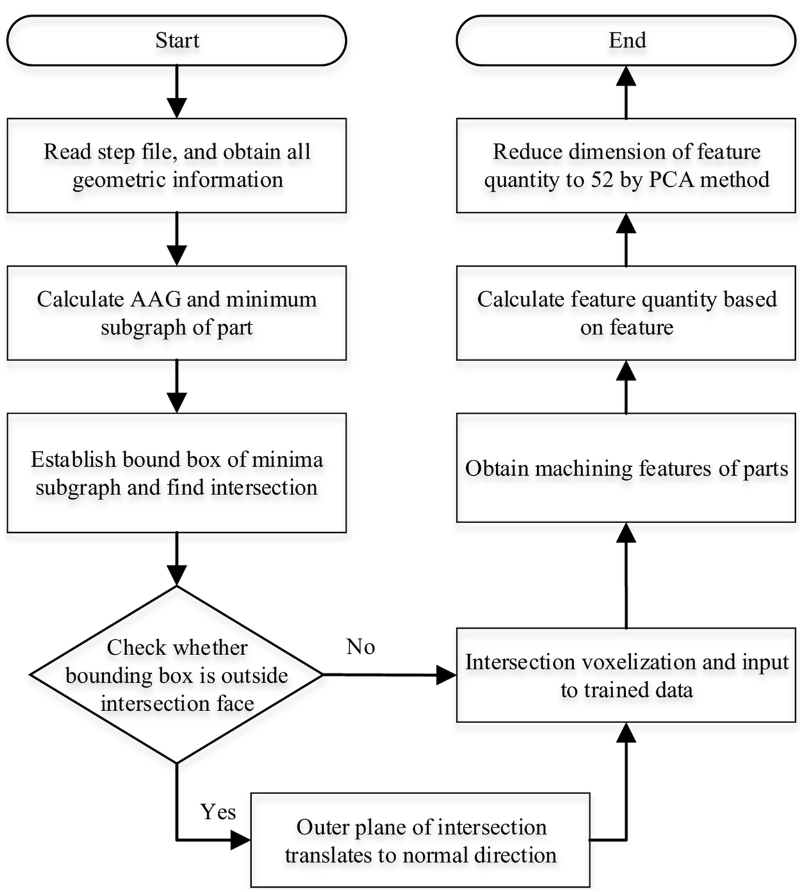


Fig. 10. Process of feature data generation.

5. Cost estimation

5.1. SVM method

The SVM is a type of generalized linear classifier, which categorizes data according to the supervised learning method. Its decision boundary is the maximum margin hyperplane [36,37]. The SVM improves the generalization ability by seeking structural risk minimization to reduce risk and confidence range.

For linear functions:

$$f(x) = kx + b \tag{4}$$

$\{x_i, y_i\}$, $i = 1, 2, \ldots, m$, $x_i \in R$, $y_i \in R$. If $f(x)$ can estimate all data with an accuracy of ε, then the problem of minimum k is a convex optimization, i.e., $min\frac{1}{2}\|k\|^2$.

By using the relaxation variables, ξ^i and ξ^{i*}, the optimization objectives are obtained.

$$min\frac{1}{2}\|k\|^2 + C\sum_{i=1}^{l}(\xi^i, \xi^{i*}) \tag{5}$$

The constraints are as follows: by applying the Lagrange function and dual variable, the dual problem is obtained.

$$f(x) = \sum_{i=1}^{l}(\alpha_i - \alpha_i^*)K(x_i, x) + b \tag{6}$$

As a kernel function, $K(x_i, x)$ satisfies the Mercer condition and $K(x_i, x) = (\varphi(x_i)\varphi(x_j))$. The radial basis function (RBF) is a universal kernel function.

$$K(x_i, x') = exp(-\|x - x'\|^2/\sigma^2) = exp(-\gamma\|x - x'\|^2) \tag{7}$$

In, $\gamma = 1/\sigma^2$, $\sigma > 0$ is the nuclear width coefficient. The selection of the penalty coefficient (C), insensitivity coefficient (ε), kernel function, and related parameters has a considerable influence on the effect of SVM.

5.2. BP neural network

The BP neural network is a multilayer feedforward neural network, which is characterized by the forward feedback of signal and reverse transmission of error. In the forward transmission, the input signal is processed layer by layer from the input layer to the output layer [38,39]. The state of neurons in each layer only affects the neurons in the next layer. If the output layer cannot yield the expected output, back propagation is employed to adjust the network weight and threshold according to the error so that the predicted output of the BP neural network would approach the output. The topology is shown in Fig. 11.

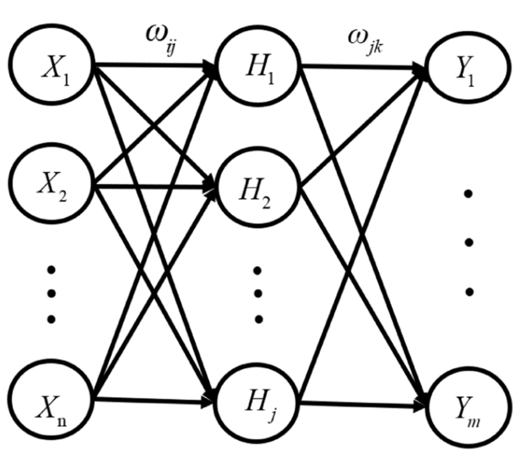


Fig. 11. Structure of BP neural network.

The input value, the predicted value, and the weight of the BP neural network are $X_1, X_2, \ldots, X_n$, $Y_1, Y_2, \ldots, Y_n$, and (ω_{ij}, ω_{jk}), respectively. The hidden layer output (H) is calculated according to the input (X), the connection weight (ω_{ij}) between the output layer and hidden layer, and the hidden layer threshold (a).

$$H_j = f\left(\sum_{i=1}^{n}\omega_{ij}x_i - a_j\right), j = 1, 2, \ldots, m \tag{8}$$

The number of nodes in the hidden layer is j, and f is the excitation function in the hidden layer that has a variety of expressions in this study.

$$f(x) = \frac{1}{1 + e^{-cx}} \tag{9}$$

According to the hidden layer output (H), connect the weight (ω_{jk}) and threshold (b), and calculate the BP neural network prediction output.

$$O_k = \sum_{j=1}^{l}H_j\omega_{jk} - b_k, k = 1, 2, \ldots, m \tag{10}$$

According to the network prediction output (O) and expected output (y), calculate the network prediction error (e).

$$e_k = Y_k - O_k, k = 1, 2\ldots, m \tag{11}$$

Update the network connection weight (ω_{ij}, ω_{jk}).

$$\omega_{ij} = \omega_{jk} + \eta H_j(1 - H_j)x(i)\sum_{k=1}^{m}\omega_{jk}e_k \tag{12}$$

$$\omega_{jk} = \omega_{jk} + \eta H_j e_k \tag{13}$$

Note that $i = 1, 2, \ldots, n$; $j = 1, 2, \ldots, l$; $k = 1, 2, \ldots, m$; and η is the learning rate. The network threshold (a, b) is thereafter updated according to the network prediction error, e.

$$a_{ij} = a_{ij} + \eta H_j(1 - H_j)\sum_{k=1}^{m}\omega_{jk}e_k \tag{14}$$

$$b_k = b_k + e_k \tag{15}$$

6. Experiments

In this study, six types of common parts are examined. Some of these, including guide shaft, guide-shaft bearing, metal gasket, fixing ring, roller, and cantilever pin are listed in Table 5. These parts contain cylinder, decahedron, through hole, and hexahedron. The training data in this study include the processing features of these parts.

6.1. Processing feature recognition

6.1.1. Training process

In this paper, 3D CNN is used to train the processing feature data of parts, to realize the processing feature recognition of parts. In this study, Google's deep-learning framework TensorFlow was used. The experiment was conducted using the Ubuntu 18.04 operating system, and the computer had the following specifications: an i9 graphics processor, GeForce 2080Ti memory (11 GB), and a 1.2-TB hard disk. The CNN data flow was designed in TensorFlow using Python. The dataset was split into three subsets: training, validation, and testing datasets (60 %, 20 %, and 20 %, respectively). The model was trained using the training dataset and validated using the validation dataset. The specific amount of data is shown in Table 6.

In this study, 3D CNN is employed to train the processing feature data of parts to achieve part-processing feature recognition. Table 7 lists the network structure parameters and training hyperparameters in the experiment.

Table 5
Some parts in the 6 classes.

Guide shaft	Guide-shaft bearing	Metal gasket	Fixing ring	Roller	Cantilever pin
					

Table 6
Training data.

Data sets	Amount of data
Training data	12,384
Validation data	4128
Testing data	4128

Table 7
Network structure parameters and training hyperparameters.

Structure	Layer (type)	Output shape
Convolution	Con2d_1	1,646,432
	Con2d_2	1,646,432
	Con2d_3	1,323,232
	Con2d_4	1,323,264
	Con2d_5	1,161,664
Fully connected	dense	124
	dense	124
	dense	14

This study also uses Adam optimizer, and the rectified linear unit (ReLu) is employed as the activation function. The initial learning rate is set to 0.001, and after 223 epochs, the training ends. Fig. 12 shows the training process of 12,384 part processing features. Based on the figures, it can be concluded that the network`s loss value decreases with iteration, and its classification accuracy improves. The accuracy of the training set and verification set is increasing, and no overfitting phenomenon is observed. The learned model produces training and validation accuracies of 98.31 % and 98.75 %, respectively. The accuracy of the test dataset obtained using the learned model is 97.70 %.

6.1.2. Analysis of results

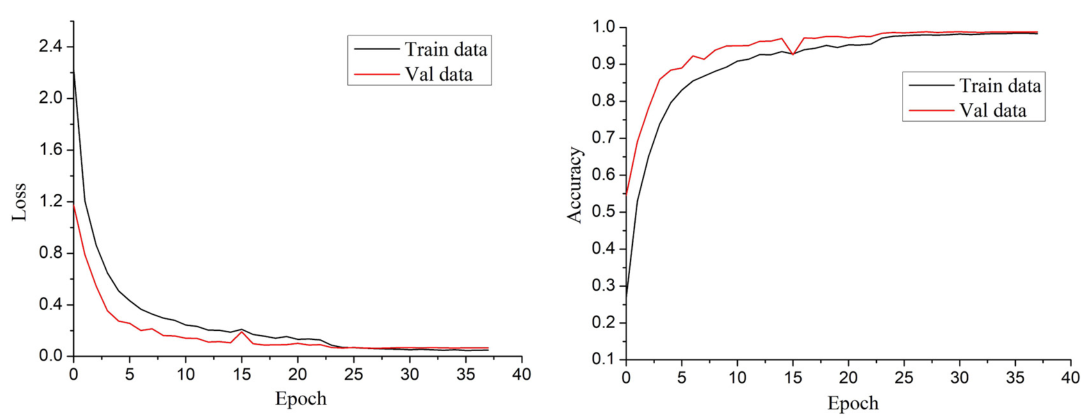
As presented in Fig. 13, the considered part contains 20 processing features. The method proposed in this paper recognizes 20 features in less than 1 s, and all of them are correctly identified; this was compared to the method in Ref. [40] that presented a new hybrid (graph + rule-based) approach for recognizing features. The method in Ref. [40] requires 2 s to identify these features, and features 5, 9, 13, and 16 are identified as stepped holes, which in Ref. [40] are defined as a series of coaxial holes, such as counter bores or counter-sunk holes. In Ref. [40], the features are defined by graphs and rules, which require complex definitions, and a long time is needed to search for specific features.

6.2. Part manufacturing cost estimate

After feature recognition, the part feature quantity is calculated, and after dimension reduction by the PCA method, 52 unidimensional data that directly or indirectly affect the part cost are generated; the influence relationship may be linear or nonlinear. In order to estimate the manufacturing cost of parts, the SVM and BP neural network are employed to study these data. In the experiment, 21,943 CSV files that contain data are generated from 21,943 step files. These data are divided into training set, verification set, and test set; the quantities are listed in Table 8. For the BP neural network, the dataset is split into three subsets: training, validation, and testing datasets (60 %, 20 %, and 20 %, respectively). The model is trained and validated using the corresponding dataset. For the SVM, the dataset is split into two subsets: training and testing datasets (80 % and 20 %, respectively). The model is trained using the former dataset. All models are tested by the same testing dataset.

6.2.1. SVM method

The software packages of SVM include LibLinear and LibSVM. In this study, ε -SVM is employed to study the cost of parts. In order to avoid the influence of large differences among input data on the training effect, this study normalizes the data with a normalization interval of $[-1,1]$. The RBF kernel function is utilized in this study. The grid search method is employed to identify the group with the best (C, γ), and the five-fold cross-validation test parameters are selected through grid search (large-scale parameter search) and fine grid search (small-scale search). The contour map is obtained after the search, as shown in Fig. 14. By means of the search, the best parameters (C $=4$



(a) Convergence of the loss function **(b) Convergence of accuracy function**

Fig. 12. Training process of processing features.
(a) Convergence of the loss function. (b) Convergence of accuracy function.

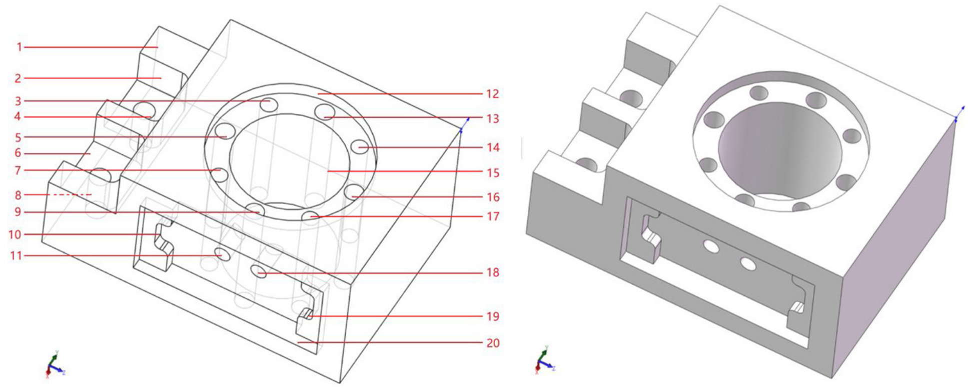


Fig. 13. 3D model of literatures [40].

Table 8
Data sets for SVM and BP neural network.

Data sets	SVM	BP neural network
Training data	17,553	13,165
Validation data	\	4338
Testing data	4338	4338

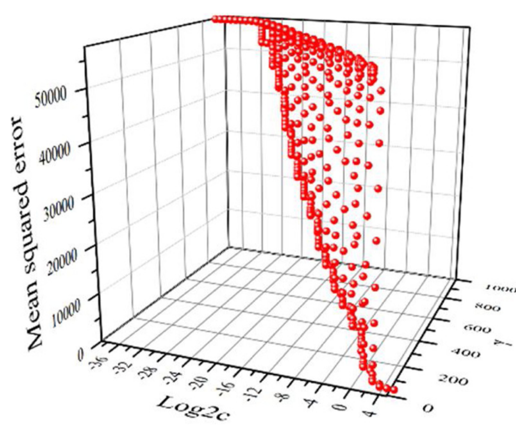


Fig. 14. Optimization diagram of SVM parameters.

and $\gamma = 0.02$) are obtained. In this study, the insensitivity coefficient, $\varepsilon = 0.01$, is selected after considering the calculation amount and accuracy.

After 8998 iterations, the constant term is found to be $b = -437.24$, the number of support vectors is 16,571, and the number of support vectors on the boundary is 16,560. The prediction of the model is that the mean square deviation is 1270.70, and the correlation coefficient is 0.906. It is not easy to determine whether the relationship between the part feature quantity and cost estimation is linear or nonlinear; therefore, LibLinear is first utilized. The software package LibLinear is based on LibSVM, which is suitable for large-scale data and high-dimensional sparse feature classification and regression. The linear classifier is mainly used to realize millions of data and features. In the experiment, the same parameters ($C = 4$ and $\gamma = 0.02$) are used. After five iterations, the operation stops. The prediction results of the model are as follows: the mean square deviation is 1167.76, and the square correlation coefficient is 0.956.

6.2.2. BP neural network method

Google`s deep-learning framework, TensorFlow, is employed in this study. In order to avoid the influence of large differences among input data on the training effect, data is normalized with a normalization interval of [0,1]. The BP network structure consists of three layers, and the initial learning rate is set as 0.001. The activation function is ReLu, and the optimizers are RMSProp and Adam. The principle of the RMSProp optimizer is similar to that of the momentum gradient descent algorithm, which can take larger steps in the horizontal direction and converge faster. The updating of parameters by Adam optimizer is not affected by the scaling transformation of gradient. It is therefore extremely suitable for large-scale data and parameter scenarios.

Fig. 15 shows the convergence of loss functions for training. It can

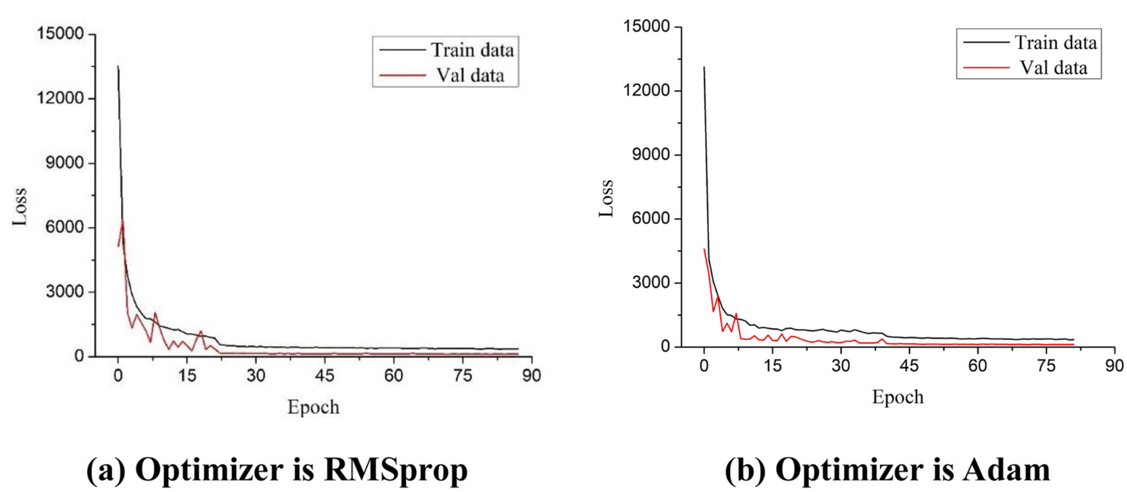


Fig. 15. Convergence of the loss functions for training.
(a) Optimizer is RMSprop. (b) Optimizer is Adam.

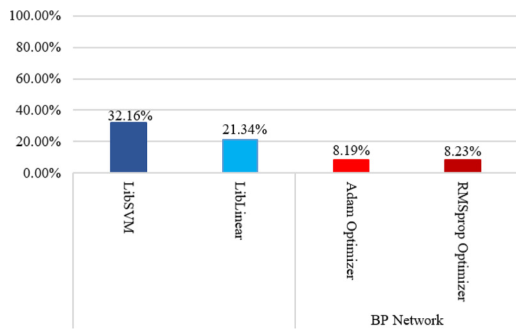


Fig. 16. *MAPEs* for different methods.

be concluded that losses in the training and verification sets decrease and tend to be stable as the iterations increase, indicating that the BP network has a better learning ability. The accuracy of the training set and verification set is increasing, and no overfitting phenomenon is observed. By comparing the loss function iterations under different optimizers, it can be observed that the convergence trend, convergence time, and training process are practically the same.

To measure accuracy, the MAPE, which avoids the problem of mutual cancellation of errors, is adopted. As shown in Fig. 16, it can accurately reflect the magnitude of actual prediction errors. The errors are calculated from the testing dataset.

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \tag{16}$$

Here, y_i represents the real price of parts, and $\hat{y}_i$ represents the predicted price of the trained model. Fig. 16 presents the *MAPE* values for LibLinear, LibSVM, and BP Network.

It can be concluded from the *MAPE* values that LibLinear method yields smaller values than the LibSVM. The method also has a better prediction effect and shorter training time. In addition, the *MAPE* value of BP neural network is significantly smaller than those of LibLinear and LibSVM. It shows that for large amounts of data, the learning effect of BP neural network is better than those of LibLinear and LibSVM. The *MAPE* values of the two training methods of RMSprop and Adam optimizer are used. The training time is similar, indicating that the BP neural network is more accurate for establishing the regression relationship between feature quantity and part cost.

7. Conclusion

In order to realize the cost estimation of parts, the present study proposes a creative method for using 3D CNN to solve the problem of feature recognition of part processing and achieves great results. Compared to other methods, the results demonstrated that the proposed method has high accuracy and high recognition speed. Moreover, it also has good expansibility by completing the training data of new features, which can identify new features.

To estimate the cost of parts, this paper describes the part-processing features in terms of feature quantity, which is also calculated. The PCA method is employed to reduce the feature quantity dimensions and obtain the training data that contain the feature quantity. The LibLinear, LibSVM, and BP network methods are utilized to study the relationship between data and cost. Thereafter, the three methods are compared according to the MAPE values. It is concluded that the estimation effect of LibLinear is slightly better than that of LibSVM. The optimizers of the BP network, Rmsprop and Adam, exhibit excellent performance for the regression of part cost estimation. The BP network has the best estimation effect and therefore has considerable potential for applications.

The cost estimation method proposed in this paper can accurately estimate parts with a complex structure, as long as more part machining feature training data are added in the actual application. The cost of this type of part is mainly focused on the part structure. Some parts, necessitate certain surface features, such as roughness, and geometric tolerances, which may be mandated by specific requirements. The part cost method proposed in this paper, however, does not recognize the aforementioned features. Future research will therefore focus on recognizing information related to such processing requirements.

In addition, the estimated cost can be used to analyze the cost of each feature in the part and guide the part design, which will be a valuable application. Our future work is to realize the visualization of the neural network, study the weight of feature quantity, and establish the relationship between cost and feature quantity.

Declaration of competing interest

None.

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